

Module13 – Visual Design Critique and Iteration

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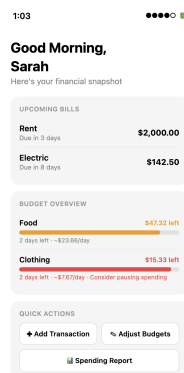
Previous URL (“after” for module 10):

<https://laws-smith-luna.github.io/User-Interface-Design/after-m10/>

After URL: <https://laws-smith-luna.github.io/User-Interface-Design/after-m13/>

Weakness #1: No budget aggregate overview

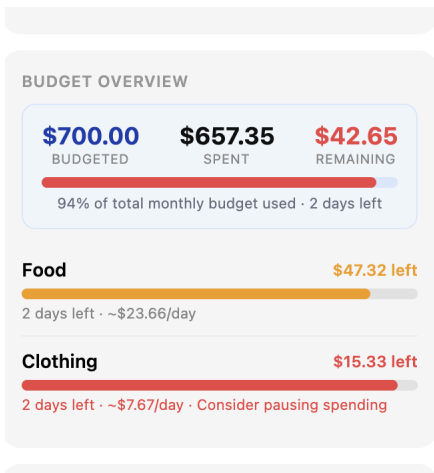
Screenshot of the issue:



Explanation of the issue:

In the information visualization reading we learn that information visualizations should provide an overview of the information to users then allow users to filter the information for specific tasks. In its current form, our app does not provide an overview of the user's budget. It displays each individual budget, but not total spending and total money remaining to be spent.

Screenshot of the fix:

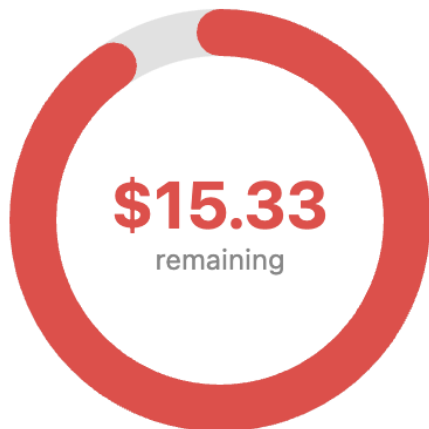


Explanation of the fix:

The new budget overview section now displays total budget, spend, and remaining budget. Now users can quickly see the overall state of their finances and look at individual budgets if they want more information.

Weakness #2: Donut charts do not have alert threshold indicator

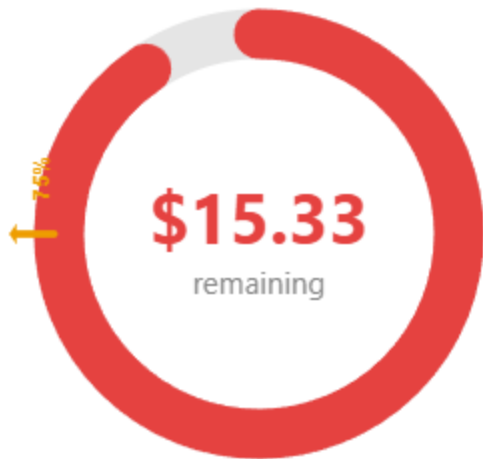
Screenshot of the issue:



Explanation of the issue:

The donut charts displaying how much of the budget the user has used so far does not have an indicator for the alert threshold the user set during the creation of the budget. In the mapping data to physical form screencast we learn that marks should be used in visual structures to communicate important information. In our donut chart, we do not have a mark to indicate the alert threshold for the budget.

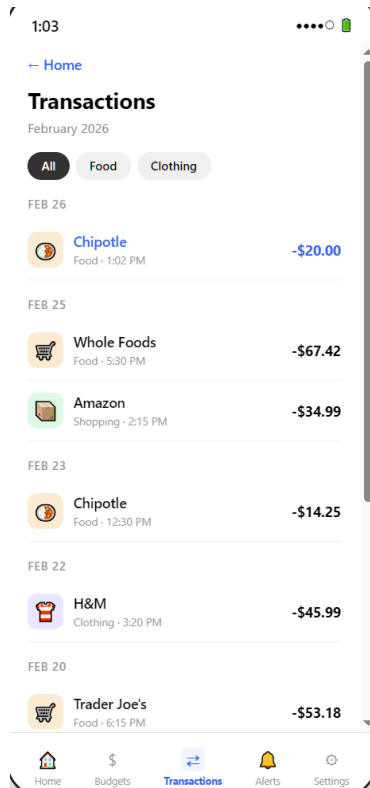
Screenshot of the fix:



There is now an indicator on the donut chart where users set their alert threshold. Now users are able to quickly determine where their spending is in relation to their set alert threshold. This means users won't be taken by surprise when that threshold is reached.

Weakness #3: No transactions filtering

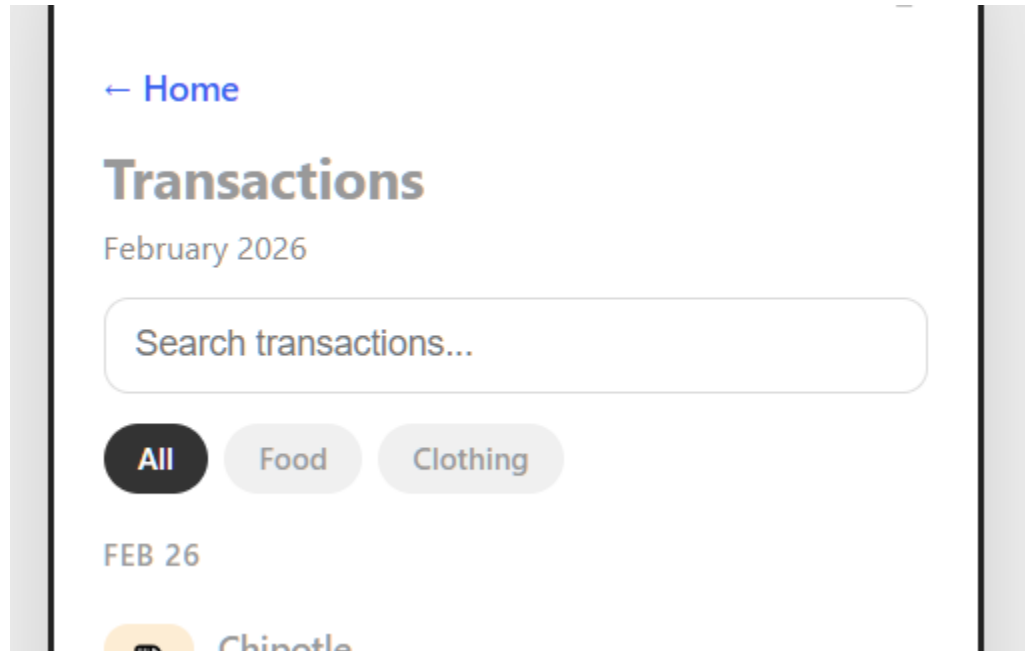
Screenshot of the issue:



Explanation of the issue:

Users are able to see all their transactions for the month, but they are unable to search for specific transactions. In the visual information module, the process of filtering information for specific tasks is emphasized, and this filtering is not possible with our current transactions page.

Screenshot of the fix:

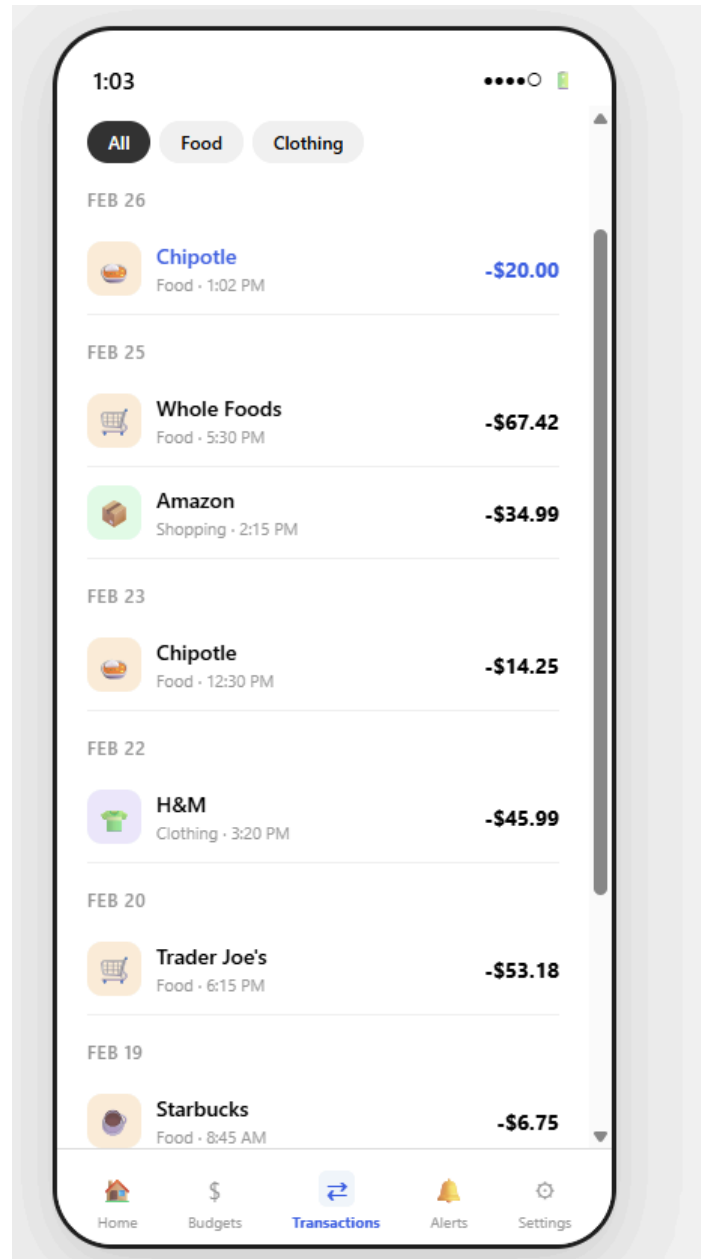


Explanation of the fix:

There is now a search bar at the top of the transactions page. Users now have the option to search for a specific transaction. This fulfills the workflow outlined in the module where users want to see a general overview of information with the ability to filter the information based on their specific task.

Weakness #4: Transaction amounts are encoded only as text, with no visual size comparison

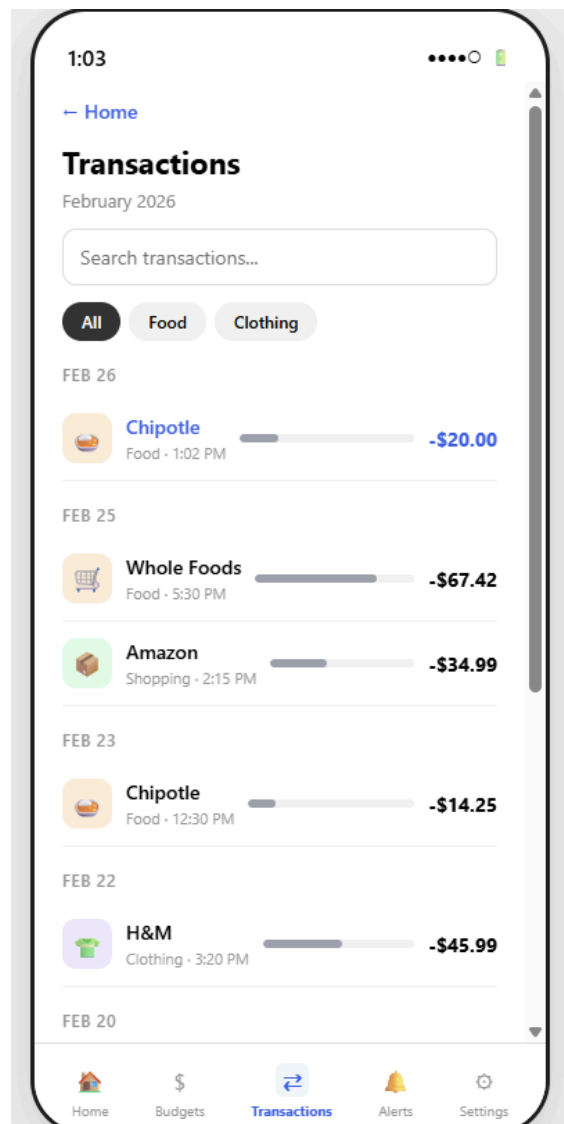
Screenshot of the issue:



Explanation of the issue:

The Transactions screen displays every transaction for the month as a row with the dollar amount printed as text on the right side. To identify which purchases drove the most spending, the user has to read every number and compare them mentally. In Module 12's Mapping Data to Physical Form lecture, position and size are named as the two most effective graphical properties for encoding quantitative data, and Tufte's principles of graphical excellence say a good visualization should "encourage the eye to compare different pieces of data." A column of text numbers does not use either property, so the screen forces the user to do the comparison work that the display itself should be doing for them.

Screenshot of the fix:

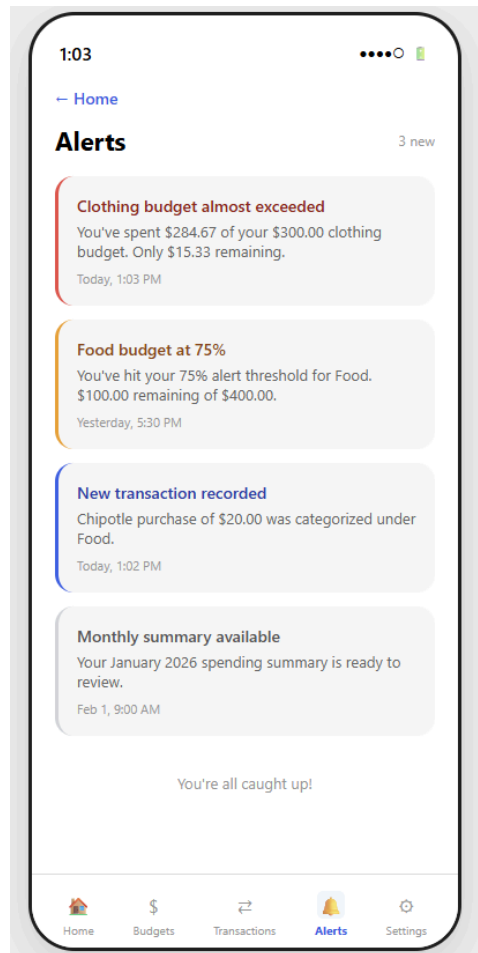


Explanation of the fix:

The updated transactions list now draws a light gray horizontal bar behind each amount, scaled so the largest transaction of the month fills the full width and every other row is a proportional fraction of that width. Users can glance down the column and see immediately that Nike.com was the biggest single purchase of the month, that Whole Foods was next, and that Starbucks was the smallest. The precise dollar amount still shows in text for users who want the exact number, but perception does the comparison work that reading cannot.

Weakness #5: Alert cards encode severity only through border color, with no text label or legend

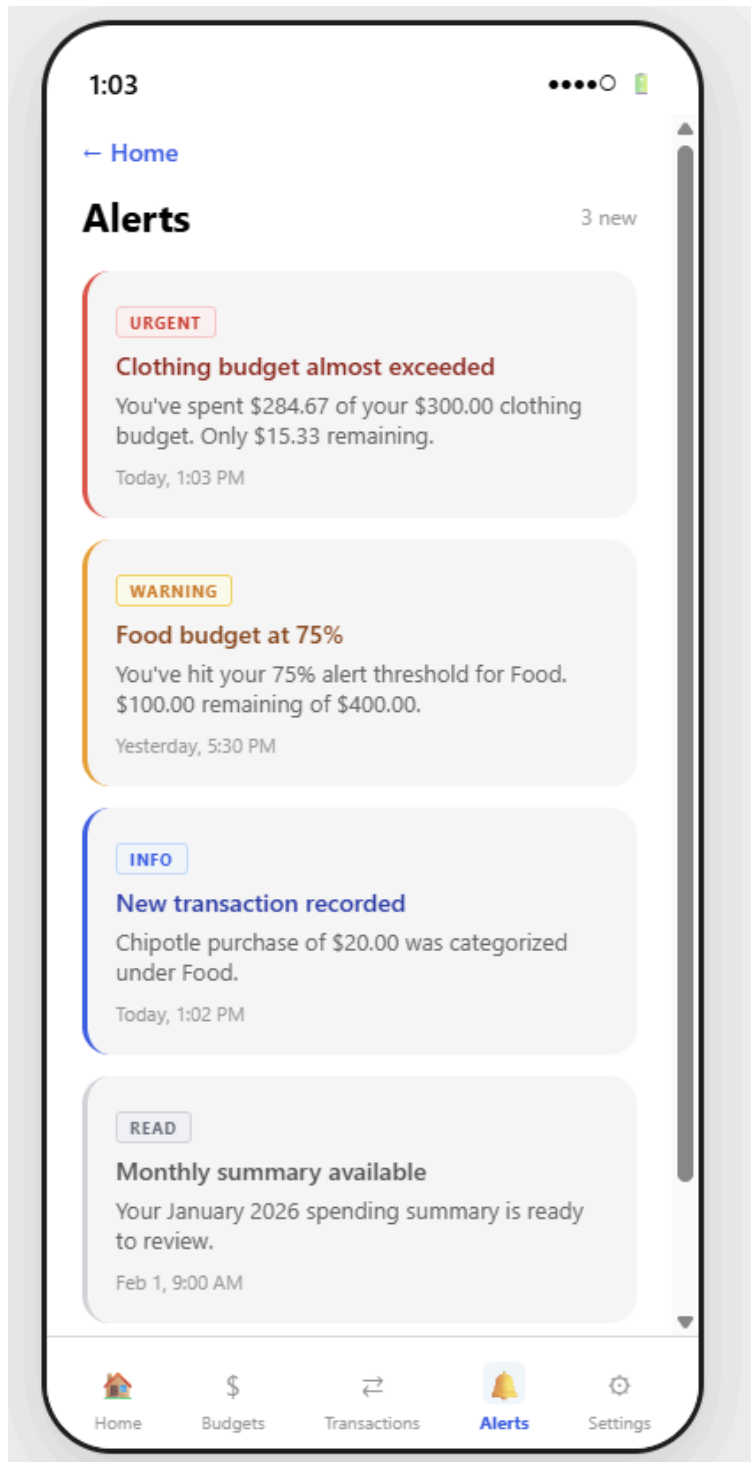
Screenshot of the issue:



Explanation of the issue:

The Alerts screen shows four cards with color-coded left borders: red for "Clothing budget almost exceeded," yellow for "Food budget at 75%," blue for "New transaction recorded," and gray for "Monthly summary available." The colors are meant to encode severity, but nothing on the screen explains what the colors mean. The user has to infer from each card's text that red is urgent, yellow is a warning, blue is informational, and gray is already handled. Module 13's design heuristics say to label visual encodings directly rather than rely on a legend, and in chapter 11 of *Don't Make Me Think* Krug points out that color-only encoding fails for users with color vision deficiency, who may not be able to distinguish the red and gray borders at small sizes.

Screenshot of the fix:



Explanation of the fix:

Each alert card now has a small uppercase tag in the top-left that names the severity directly: "URGENT" in red, "WARNING" in amber, "INFO" in blue, and "READ" in gray. The tag sits right next to the colored border, so users who can see the colors get reinforcement and users who

cannot still get the severity straight from the tag text. Severity is now encoded in both color and text, which means a colorblind user can triage the alert list as quickly as anyone else.

Weakness #6: Important alerts not visually dominant enough

Principle violated: Visual hierarchy / Attention

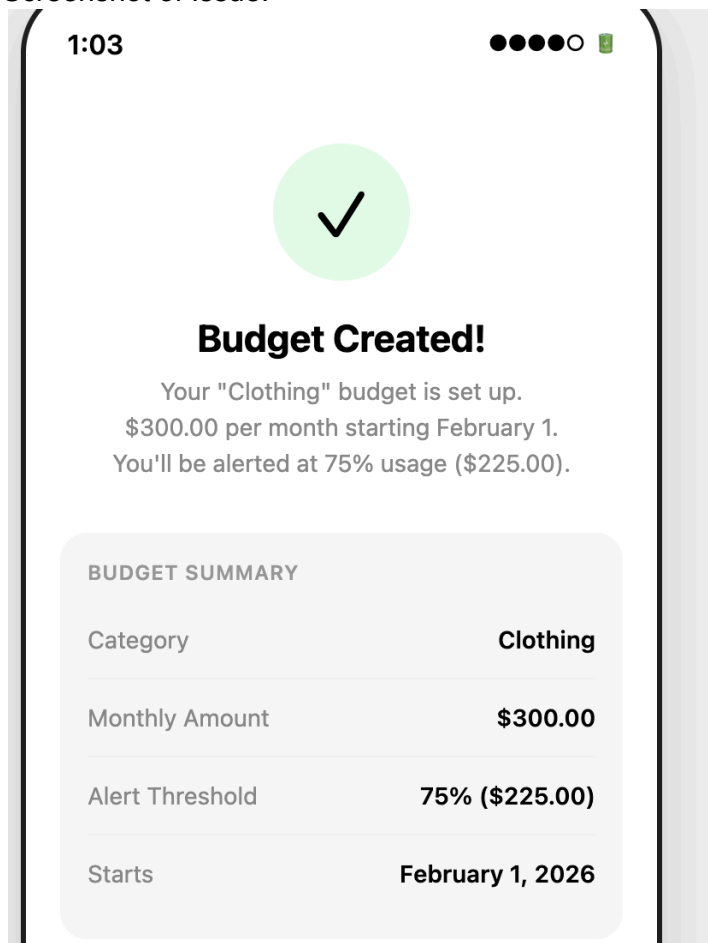
Problem:

Critical warnings don't stand out strongly.

Problem (Explanation of Issue)

The alert messages in the interface do not stand out sufficiently from surrounding content, making it difficult for users to quickly identify critical information. Important alerts, such as low budget warnings, visually resemble standard cards and lack strong emphasis. This violates the principle of visual hierarchy, which states that more important information should be more visually prominent. Because the alerts are not clearly differentiated through size, contrast, or emphasis, users may overlook urgent messages, reducing the effectiveness of the system in supporting timely decision-making.

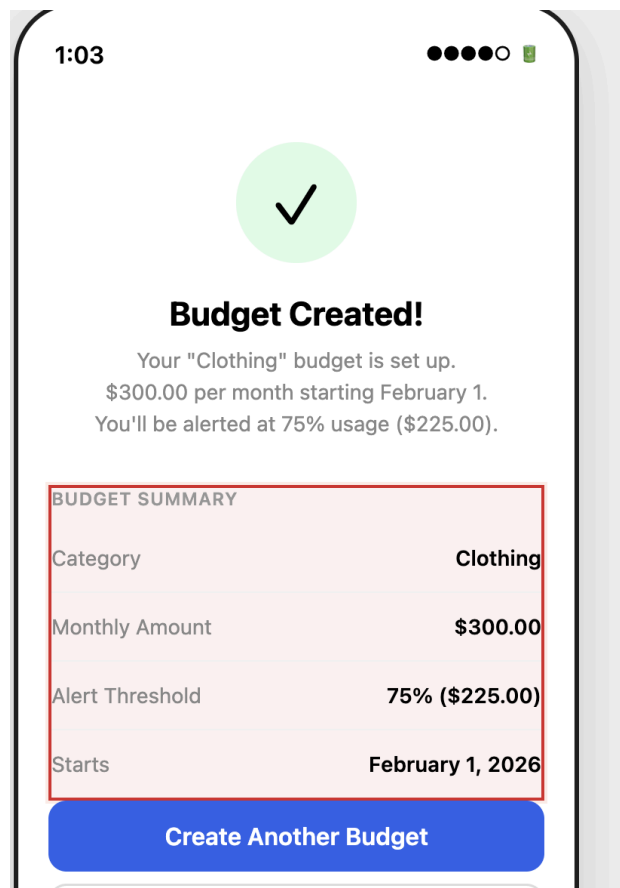
Screenshot of Issue:



Explanation of Fix:

To address this issue, the design enhances the visual prominence of important alerts by increasing contrast, adding stronger borders, and introducing subtle background highlighting. For example, critical alerts now use a thicker red border, a lightly tinted background, and additional visual emphasis such as bold text or shadow. These changes create a clear distinction between standard content and high-priority alerts, aligning with visual hierarchy principles. As a result, users can more easily identify and prioritize urgent information at a glance.

Screenshot of fix:



Weakness #7:Color contrast issues

Principle violated: Accessibility (WCAG contrast)

Problem:

Light gray text (e.g., #999) is hard to read.

Problem (Explanation of Issue)

Several text elements in the interface use very light gray colors (e.g., #999), which reduces readability, especially for users with visual impairments or when viewed on lower-quality screens. This violates accessibility guidelines, particularly WCAG contrast standards, which recommend sufficient contrast between text and background to ensure readability. Poor contrast makes it harder for users to scan information efficiently and can lead to missed or misunderstood content.

Screenshot of issue:

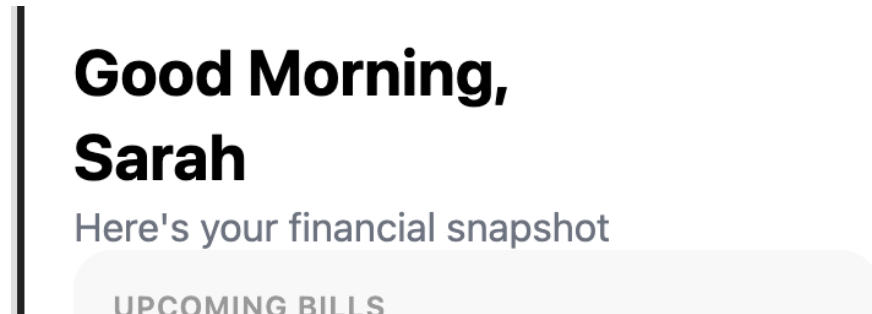


**Good Morning,
Sarah**
Here's your financial snapshot

Explanation of Fix:

The issue is addressed by updating text colors to darker, higher-contrast values (e.g., replacing #999 with #6b7280). This improves legibility while maintaining the overall visual style of the interface. By increasing contrast, the design becomes more accessible and easier to read for a wider range of users. This change aligns with accessibility best practices and enhances overall usability without significantly altering the layout.

Screenshot of Fix:



**Good Morning,
Sarah**
Here's your financial snapshot

UPCOMING BILLS

Weakness #8: Cards lack depth (flat design)

Principle violated: Visual hierarchy / affordance

Problem:

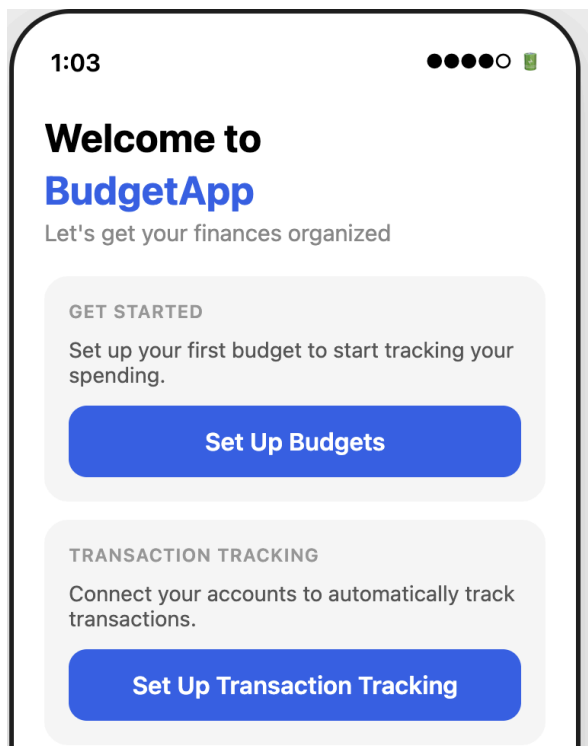
All cards look flat and equal, making it hard to distinguish importance.

Explanation of Issue:

The card components in the interface appear visually flat, with minimal distinction between different layers of content. This makes it difficult for users to perceive grouping and hierarchy within the layout. The lack of depth violates principles of visual hierarchy and layering, where subtle visual cues such as shadows or elevation help users understand which elements are interactive or grouped together. Without these cues, the interface can feel cluttered and harder to navigate.

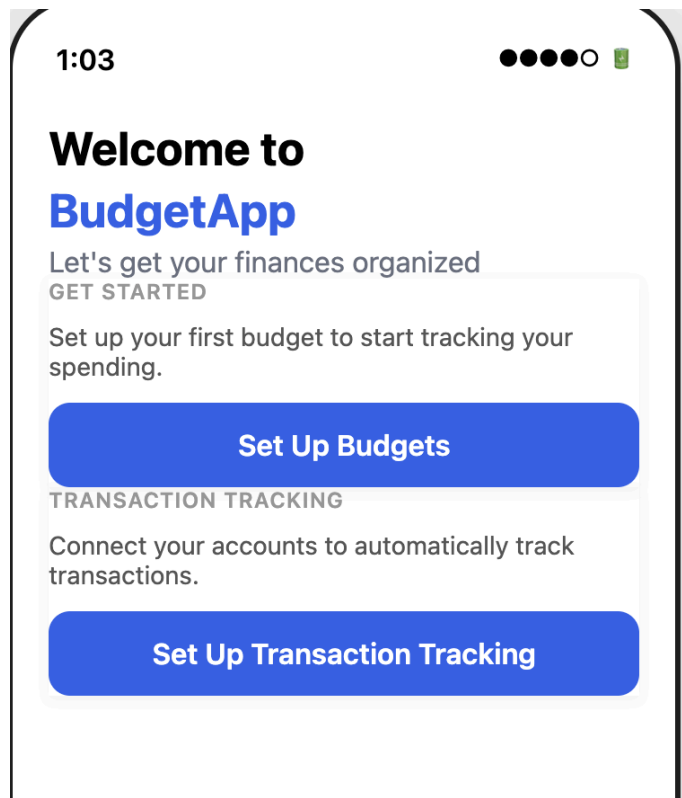
Fix (CSS):

Screenshot of issue:

**Explanation of Fix:**

To improve this, subtle shadows and elevation effects are added to card components, along with hover states that increase shadow depth slightly. These changes create a clearer sense of layering and separation between elements. By introducing depth, the interface becomes more visually organized and easier to scan. This aligns with modern design practices and improves both usability and aesthetic clarity without overwhelming the user.

Screenshot of fix:



Weakness #9: No clear clickable indication on cards

Principle violated: Affordance

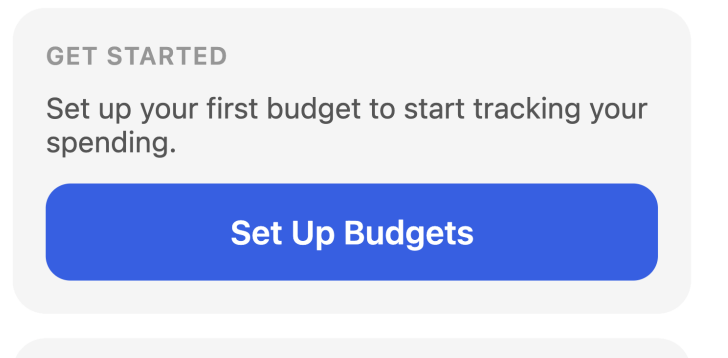
Problem:

Clickable cards don't look interactive.

Explanation of Issue:

Several card elements in the interface are interactive (e.g., budget cards that navigate to detail views), but they do not clearly communicate that they are clickable. Visually, these cards appear identical to static content, lacking cues such as hover states, cursor changes, or visual affordances. This violates the principle of affordance and signifiers, which states that interactive elements should clearly indicate how users can interact with them. As a result, users may not realize that the cards are actionable, leading to missed functionality and reduced usability.

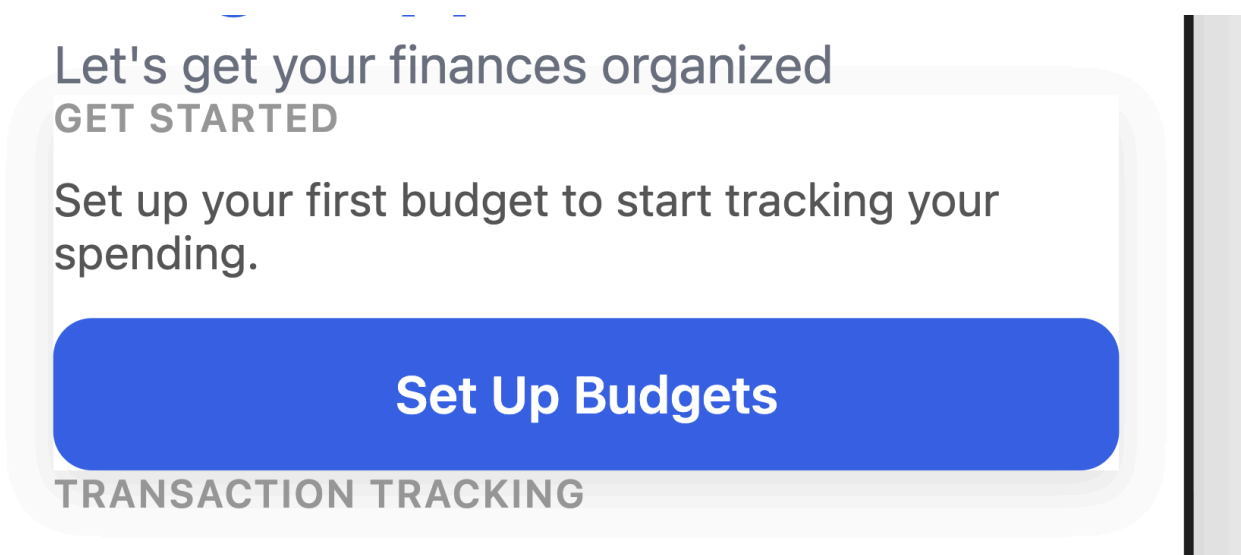
Screenshot of Issue:



Explanation of Fix:

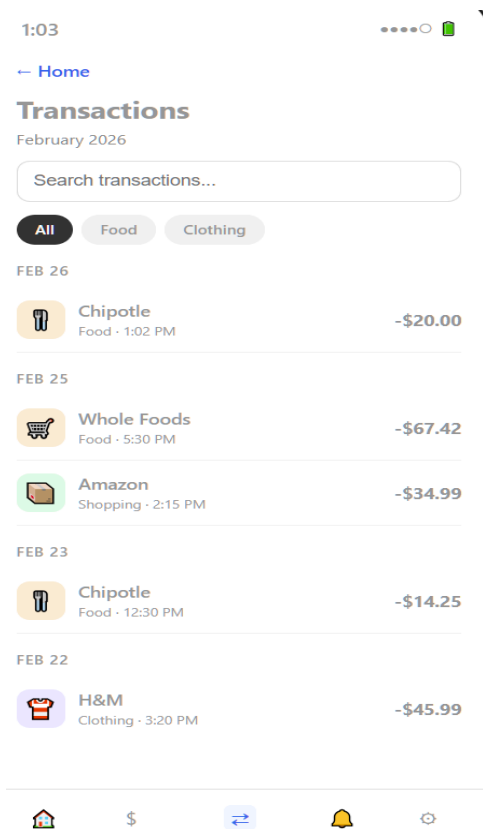
To address this issue, visual signifiers are added to clearly indicate interactivity. Clickable cards now include a pointer cursor on hover, a subtle background color change, and/or a slight elevation effect to signal responsiveness. These cues provide immediate feedback that the element can be interacted with, aligning with affordance principles. By making interactivity more visible, users can more easily discover navigation options and engage with the interface confidently.

Screenshot of fix:



Weakness #10: No way of looking at transactions from previous month

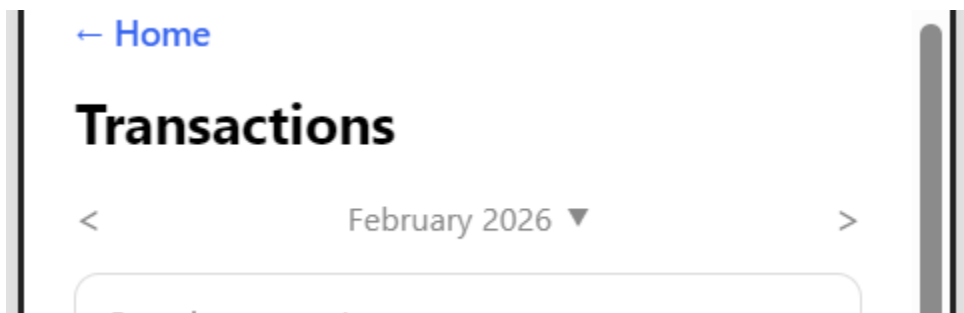
Screenshot of the issue:



Explanation of the issue:

The transactions page only shows the current month's transactions. There is no way to view previous transactions. We learned that information visuals should provide an overview as well as the ability to filter information. In its current state, the transactions page does not give an overview or the ability to filter through information from previous months.

Screenshot of the fix:



Explanation of the fix:

The transactions page now has the ability to show previous month's transactions. This allows users to get a complete overview of their spending as well as filter information within a certain month.